

The University of Azad Jammu and Kashmir, Muzaffarabad.



Computer Networks

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LAB 7 — CLI COMMANDS UNDERSTANDING IN CISCO PACKET TRACER

1. Objective

To practice and understand basic Cisco IOS Command Line Interface (CLI) commands used to access a router or switch, enter configuration modes, configure interfaces, and verify network-device status and configuration.

2. Introduction

The Cisco IOS CLI provides a text-based method for operating and configuring Cisco network devices. Commands are entered in different modes, including user EXEC, privileged EXEC, global configuration, and interface configuration mode. The following commands represent the basic CLI operations used during router access and configuration work.

3. Common CLI Commands

Command	Purpose
enable	Enters privileged EXEC mode.
configure terminal	Enters global configuration mode.
hostname R1	Changes the device hostname to R1.
interface gigabitEthernet 0/0	Enters configuration mode for the selected interface.
no shutdown	Enables an administratively disabled interface.
ip address 192.168.10.1 255.255.255.0	Assigns an IPv4 address and subnet mask to an interface.
exit	Moves back one configuration level.
end	Returns directly to privileged EXEC mode.
show ip interface brief	Displays interface IP addresses and operational status.
show running-config	Displays the current running configuration.
copy running-config startup-config	Saves the current configuration to startup configuration.

4. Basic CLI Procedure

1. Open the device in Cisco Packet Tracer and select the CLI tab.
2. Enter privileged mode using enable.
3. Enter configuration mode using configure terminal.
4. Select the required interface with an interface command.
5. Apply the required interface configuration, such as an IP address and no shutdown.
6. Use exit or end to leave configuration mode.
7. Use show commands to verify the configuration and interface status.
8. Save the configuration using copy running-config startup-config.

5. Example CLI Configuration

```
R1> enable
R1# configure terminal
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ip address 192.168.10.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# end
R1# show ip interface brief
R1# copy running-config startup-config
```

The command show ip interface brief can be used to verify whether interfaces are up or down and whether an IP address has been assigned. The show running-config command can be used to inspect the active configuration.

7. Result

Basic Cisco IOS CLI commands were practiced for device access, interface configuration, verification, and saving the configuration.

8. Conclusion

The lab provided practical familiarity with Cisco CLI modes and commonly used commands for basic network-device configuration and verification.